Christopherson Tribonacci Projection

Nine Core Theorems with Full Proofs

A Cubic-Field-Rooted Algebraic Family

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Abstract

This document presents nine theorems constituting the core mathematical framework of the Christopherson Tribonacci Projection (CTP). The framework establishes a coherent algebraic family rooted in the cubic number field $\mathbf{Q}(\rho)$, where ρ is the tribonacci constant. Theorems T1–T4 develop the algebraic engine: a base field theorem, a universal reduction theorem, a disguised-cubic lifting mechanism, and a degree law conditional on a full-extension hypothesis. Theorem T5 proves a strict positivity result for a structural surplus function. Theorems T6–T7 establish a second algebraic family arising from the phi-point equation. Theorem T8 identifies an exceptional algebraic constant of degree 12 lying outside the regular family. Theorem T9 resolves the open condition in T4: applying a classical irreducibility criterion (Capelli, 1898) to the cubic field $\mathbf{Q}(\rho)$, it provides an exact, checkable characterization of when radical collapse occurs, transforming the conditional degree law into a complete degree formula. A structural consequence is that collapse is *generic* in quadratic fields but *exceptional* in cubic fields—a fact that gives precise algebraic content to the dimensional distinction between the golden ratio and the tribonacci constant.

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Preface I: On the Flatness of a Sacred Number

For twenty-four centuries, the golden ratio occupied a position in mathematics that was less like a constant and more like a sacrament. Euclid gave it geometric form. Fibonacci dressed it in rabbits. Da Vinci painted it into flesh. Kepler called it a jewel of geometry. And for all that time, no one interrogated the dimension in which it operated.

The golden ratio $\varphi = \frac{1+\sqrt{5}}{2}$ is the positive root of $x^2 - x - 1 = 0$. It is quadratic. It has two roots, both real, both confined to the number line. Its self-referential property—$\varphi^2 = \varphi + 1$—is a statement about *area*: the square of the length equals the length plus unity. This is a two-dimensional identity. It describes a relationship between a line segment and the flat surface it subtends. It is, in the most literal algebraic sense, a *planar* constant.

The Fibonacci recurrence $F_{n+1} = F_n + F_{n-1}$ converges in ratio to φ, and this fact has been treated as evidence of universal growth. But recurrence is not growth. Each new Fibonacci number is the *sum* of its two predecessors—and a sum, in a finite system, is a consolidation. The two inputs are subsumed. They do not persist alongside their successor; they are *consumed by it*. The sunflower does not pack seeds at φ-spacing because the arrangement is generative. It packs them at that spacing because it is the most efficient way to *fill a finite disk*. Each seed claims territory at the expense of available space. The spiral is the geometry of optimal consumption from a bounded resource.

None of this makes φ incorrect. It makes it *incomplete*. It is a constant that describes partitioning—the ratio at which a whole divides into self-similar parts—but it says nothing about the cost of the division, the energy required to separate the parts, or the dimension in which the resulting objects actually exist. It describes *where things end up*, not what it cost to put them there. It is a pre-tax price in a universe where nothing is free.

The observation that launched this work was not a formula. It was a question: *what happens to φ when you give it a third dimension?*

The answer is that φ does not generalize into three dimensions. It is replaced by a different constant, rooted in a different polynomial, with a fundamentally different algebraic character. The quadratic $x^2 = x + 1$ becomes the cubic $x^3 = x + 1$, and the solution is no longer $1.618\ldots$ but $1.3247\ldots$ The self-referential identity shifts from area to *volume*: $\tau^3 = \tau + 1$. The cube of the length equals the length plus unity. The three-dimensional object contains itself.

But the deeper shift is in the root structure. The quadratic $x^2 - x - 1 = 0$ has two real roots: φ and $-1/\varphi$. Both persist on the real line. The cubic $x^3 - x - 1 = 0$ has three roots: one real ($\tau \approx 1.3247$) and two complex conjugates. The complex pair annihilates—they cancel each other in conjugate symmetry. The single real root *survives*.

Two roots, both real, both persisting: that is the algebra of binary consumption, where every division leaves two pieces and neither disappears.

Three roots, one real, two self-annihilating: that is the algebra of ternary construction, where the binary component destroys itself and the constructive component is the one left standing.

This distinction—between a quadratic constant confined to the plane and a cubic constant native to volume—is what motivated the examination of the tribonacci constant ρ and the algebraic family it generates. The eight theorems in this document are the result.

Preface II: The Construction Surplus and the Cost of Time

There is a difference between a number that describes a static ratio and a number that describes a process. The golden ratio is a static ratio: it relates a part to a whole at a single instant, with no reference to the energy, duration, or dimensionality of the operation that created the partition. It is a snapshot. A photograph of a cut apple, taken after the knife has already done its work, with the knife edited out of the frame.

A process requires at least one additional axis: time. And time imposes cost. Every physical division—every separation of a whole into parts—requires energy input. The bonds at the cut surface are broken. The dangling bonds represent unsatisfied states. The surface energy of the two pieces exceeds the surface energy of the original. The second law of thermodynamics guarantees that this cost is never zero and never recoverable in full.

The golden ratio ignores this cost entirely. Its defining equation $a/b = (a + b)/a$ treats the division as instantaneous, lossless, and reversible. In the physical universe, no division has any of these properties.

This observation leads to a question that is mathematical rather than physical: *when a constant is promoted from two dimensions to three—when the flat self-referential identity is replaced by a volumetric one—does a surplus appear?*

Theorem T5 of this document answers in the affirmative. The structural surplus function

$$\Delta(b) = \sqrt{b^6 + b^3 + b + 1} - b^3$$

is strictly greater than $\frac{1}{2}$ for all integer bases $b \geq 2$, and approaches $\frac{1}{2}$ asymptotically from above. The surplus never vanishes. The passage from a flat partition to a volumetric one produces a *constructive remainder*—a positive quantity that does not exist in the binary, planar formulation.

This surplus is what a linear constant cannot see. The golden ratio operates on a line. A line has no interior. A volume does. And the interior of a volume is where structure accumulates—where bonds form rather than break, where information is stored rather than forwarded, where persistence replaces consumption.

The fourth dimension—time—enters as the axis along which this cost is extracted. In the linear formulation, the cost is invisible because the operation is treated as dimensionless. In the volumetric formulation, the cost is explicit: it is the difference between the naïve linear prediction and the actual higher-dimensional result. Time does not appear as a variable in these theorems, but it is the *reason* the surplus exists. Every real construction takes time. Every real bond formation has a duration. The surplus is the algebraic shadow of that duration—the mathematical residue of the fact that building something in three dimensions cannot be reduced to a sequence of one-dimensional operations without losing information.

The theorems that follow do not make physical claims. They establish algebraic identities, field-theoretic structures, and polynomial families. But the *reason* these structures were sought—the reason anyone looked inside the tribonacci constant and asked what it contained—was the recognition that a flat number cannot describe a volumetric universe, and that the mathematics of construction must, at minimum, be cubic.

Preface III: Applied Dimensional Mathematics

The standard practice in applied mathematics is to solve problems within the dimension in which they are posed. A one-dimensional differential equation is solved on a line. A two-dimensional boundary-value problem is solved on a plane. The dimension is given by the problem, and the mathematics conforms to it.

This work proceeds in the opposite direction.

The starting observation was that certain fundamental constants—φ, π, the Fibonacci ratio—are treated as universal, but are in fact *dimensionally specific*. The golden ratio is a quadratic object: it is native to degree-two polynomials, two-dimensional self-similarity, and binary recurrence. It does not naturally extend into three dimensions. Attempting to "use φ in 3D" is not extending it; it is embedding a flat object into a volume and pretending the embedding is free. It is not.

Applied Dimensional Mathematics, as practiced in this document, begins with the dimension and derives the constant.

If the problem is inherently three-dimensional—if the objects of interest occupy volume, if the processes of interest require duration, if the structures of interest persist rather than partition—then the appropriate algebraic foundation is cubic, not quadratic. The relevant constant is not the root of $x^2 = x + 1$ but of $x^3 = x^2 + x + 1$, or $x^3 = x + 1$, or some other irreducible cubic matched to the structure under study.

The tribonacci constant $\rho \approx 1.8393$ is the real root of $x^3 - x^2 - x - 1 = 0$. Unlike φ, it generates a *cubic number field*—a three-dimensional algebraic object with basis $\{1, \rho, \rho^2\}$. Every element of this field reduces uniquely to an integer linear combination of those three basis elements. The field is closed, self-contained, and natively three-dimensional.

Within this field, a family of algebraic constants emerges. Each member is defined by a signed sum of tribonacci powers, reduced to a triple $(a, b, c) \in \mathbf{Z}^3$, lifted through iterated radical extraction, and characterized by a sparse polynomial whose degree—under a full-extension hypothesis—obeys the multiplicative law $\deg(q) = 3 \cdot 2^N$, where N is the extraction depth. The degree sequence $6, 12, 24, 48, 96, 192, \ldots$ is not empirical. It is forced by the interaction of the cubic base field with the binary radical tower, though one known collapse (the 2D subtractive member) demonstrates that the hypothesis does not hold universally.

This is what it means to do mathematics *dimensionally*: not to take a known constant and project it into a higher space, but to let the dimension of the problem determine the algebraic ground field, and then to discover what structures that field contains.

The eight theorems in this document are the first systematic output of that method applied to the tribonacci cubic. They were not obtained by generalizing φ. They were obtained by *abandoning* φ and asking what algebra looks like when the base field is cubic rather than quadratic—when the foundational identity is $\rho^3 = \rho^2 + \rho + 1$ rather than $\varphi^2 = \varphi + 1$—when the universe is assumed, from the first line, to be three-dimensional.

A note on constants. Prefaces I and II discuss the cubic $x^3 = x + 1$, whose real root $\tau \approx 1.3247$ is known as the *plastic constant*. The theorems that follow use the *tribonacci constant* $\rho \approx 1.8393$, the real root of $x^3 = x^2 + x + 1$. These are distinct algebraic numbers arising from distinct cubics. The plastic constant motivated the dimensional inquiry; the tribonacci constant is the one whose cubic field $\mathbf{Q}(\rho)$ supports the algebraic family developed in the theorems. Both are cubic irrationals, but they generate different number fields and serve different roles.

— Corick Christopherson, April 2026

1 Overview and Dependency Structure

The nine theorems are organized as follows. The preceding prefaces are motivational and interpretive; they are not used as premises in any proof.

The algebraic engine (T1–T4):

- **T1** establishes the cubic base field $\mathbf{Q}(\rho)$.
- **T2** proves that every regular CTP expression reduces to an integer-coefficient quadratic in ρ.
- **T3** shows that every such reduced element satisfies a disguised cubic, and that 2^N-th root extraction produces a sparse lifted polynomial.
- **T4** gives the degree law $\deg(q) = 3 \cdot 2^N$ under the full radical extension hypothesis.

The surplus inequality (T5):

- **T5** is analytically independent of T1–T4 and proves a strict lower bound on the structural surplus function $\Delta(b)$.

The phi-point family (T6–T7):

- **T6** establishes algebraicity of the phi-point family for algebraic parameters.
- **T7** derives the explicit degree-16 polynomial for $P = 1$.

The exceptional constant (T8):

- **T8** identifies κ_ρ as a degree-12 algebraic constant outside the regular 2^N-root branch.

The collapse criterion (T9):

- **T9** resolves the open condition in T4. Applying Capelli's irreducibility criterion to the cubic field $\mathbf{Q}(\rho)$, it provides a complete, checkable characterization of when radical collapse occurs, and proves that collapse is exceptional in cubic fields but generic in quadratic fields.

Dependency graph. T1 is foundational. T2 depends on T1. T3 depends on T1 and T2. T4 depends on T3 plus the full-extension hypothesis. T5 is independent. T6 is independent of T1–T5. T7 depends on T6. T8 depends on T1 plus explicit elimination. T9 depends on T4 plus Capelli's theorem (1898), and retroactively removes the hypothesis from T4.

Computational policy. Numerical evaluations in this document serve only as consistency checks. All theorem-level algebraic claims are established by field-theoretic arguments, explicit elimination, and irreducibility proofs over $\mathbf{Q}$.

2 T1: Tribonacci Base Field Theorem

Definition 2.1 (Tribonacci constant)**.** The *tribonacci constant* ρ is the unique positive real root of the polynomial

$$f(x) = x^3 - x^2 - x - 1.$$

Numerically, $\rho \approx 1.8392867552141611$.

Theorem 2.2 (Tribonacci Base Field)**.** *Let ρ be as in Definition 2.1. Then:*

(i) *$f(x)$ is irreducible over $\mathbf{Q}$.*

(ii) *ρ has algebraic degree 3 over $\mathbf{Q}$.*

(iii) *$\mathbf{Q}(\rho)$ is a cubic number field.*

(iv) *Every element of $\mathbf{Q}(\rho)$ can be written uniquely as*

$$u\rho^2 + v\rho + w, \qquad u, v, w \in \mathbf{Q}.$$

Lemma 2.3 (No rational root)**.** *The polynomial $f(x) = x^3 - x^2 - x - 1$ has no rational root.*

Proof. By the Rational Root Theorem, every rational root of a monic integer polynomial must divide the constant term. Here the constant term is -1, so the only candidates are $x = \pm 1$. We verify:

$$f(1) = 1 - 1 - 1 - 1 = -2 \neq 0,$$
$$f(-1) = -1 - 1 + 1 - 1 = -2 \neq 0.$$

Since f has no rational root and is cubic (hence cannot factor into two lower-degree integer polynomials without a linear factor), f is irreducible over $\mathbf{Q}$. □

Lemma 2.4 (Reduction relation)**.** *The defining relation*

$$\rho^3 = \rho^2 + \rho + 1 \tag{1}$$

holds, and every power ρ^n with $n \geq 3$ can be expressed as an integer linear combination of $\{1, \rho, \rho^2\}$.

Proof. Equation (1) follows immediately from $f(\rho) = 0$. For higher powers, one applies (1) recursively: whenever a term ρ^k with $k \geq 3$ appears, replace ρ^3 using (1) and collect terms. Since each application reduces the maximum exponent or replaces ρ^3 with lower-order terms, the process terminates in a polynomial of degree at most 2 in ρ. □

Lemma 2.5 (Basis uniqueness)**.** *Every element of $\mathbf{Q}(\rho) \cong \mathbf{Q}[x]/(f)$ has a unique representative of degree at most 2.*

Proof. The quotient ring $\mathbf{Q}[x]/(f)$ is a field (since f is irreducible) and every coset has a unique representative obtained by Euclidean division modulo the monic degree-3 polynomial f. This representative has degree at most 2. □

Proof of Theorem 2.2. Lemma 2.3 establishes irreducibility, so $[\mathbf{Q}(\rho) : \mathbf{Q}] = \deg f = 3$. This proves (i)–(iii). Part (iv) follows from Lemma 2.5: the set $\{1, \rho, \rho^2\}$ is a $\mathbf{Q}$-basis for $\mathbf{Q}(\rho)$. □

Example 2.6. Using the reduction relation repeatedly:

$$\begin{aligned}
\rho^3 &= \rho^2 + \rho + 1, \\
\rho^4 &= \rho \cdot \rho^3 = \rho(\rho^2 + \rho + 1) = \rho^3 + \rho^2 + \rho = 2\rho^2 + 2\rho + 1, \\
\rho^5 &= \rho \cdot \rho^4 = 2\rho^3 + 2\rho^2 + \rho = 4\rho^2 + 3\rho + 2, \\
\rho^6 &= \rho \cdot \rho^5 = 4\rho^3 + 3\rho^2 + 2\rho = 7\rho^2 + 6\rho + 4.
\end{aligned}$$

Every power reduces to exactly three integer coefficients.

Remark 2.7. This theorem anchors the entire CTP framework. Every subsequent construction ultimately derives its algebraic structure from the single irreducible cubic $f(x) = x^3 - x^2 - x - 1$ and the corresponding field $\mathbf{Q}(\rho)$.

Lemma 2.8 (Field generation)**.** *Since* $[\mathbf{Q}(\rho) : \mathbf{Q}] = 3$ *is prime, every element* $D \in \mathbf{Q}(\rho) \setminus \mathbf{Q}$ *generates the full field:* $\mathbf{Q}(D) = \mathbf{Q}(\rho)$*.*

Proof. The subfield $\mathbf{Q}(D)$ satisfies $\mathbf{Q} \subsetneq \mathbf{Q}(D) \subseteq \mathbf{Q}(\rho)$, so $[\mathbf{Q}(D) : \mathbf{Q}]$ divides $[\mathbf{Q}(\rho) : \mathbf{Q}] = 3$ and is strictly greater than 1. Since 3 is prime, $[\mathbf{Q}(D) : \mathbf{Q}] = 3$, and therefore $\mathbf{Q}(D) = \mathbf{Q}(\rho)$. $\square$

3 T2: CTP Reduction Theorem

Definition 3.1 (Regular CTP expression)**.** A *regular CTP expression* is any finite signed sum of non-negative integer powers of the tribonacci constant:

$$E = \sum_{j=1}^{m} \varepsilon_j \rho^{n_j}, \qquad \varepsilon_j \in \{-1, +1\}, \quad n_j \in \mathbf{Z}_{\geq 0}.$$

Theorem 3.2 (CTP Reduction)**.** *Every regular CTP expression reduces uniquely to the form*

$$D = a\rho^2 + b\rho + c, \qquad a, b, c \in \mathbf{Z}.$$

Lemma 3.3 (Integer power reduction)**.** *For each integer $n \geq 0$, there exist integers A_n, B_n, C_n such that*

$$\rho^n = A_n\rho^2 + B_n\rho + C_n.$$

Moreover, the triple (A_n, B_n, C_n) satisfies the recurrence

$$(A_{n+1}, B_{n+1}, C_{n+1}) = (A_n + B_n,\ A_n + C_n,\ A_n) \tag{2}$$

with initial conditions $(A_0, B_0, C_0) = (0, 0, 1)$, $(A_1, B_1, C_1) = (0, 1, 0)$, $(A_2, B_2, C_2) = (1, 0, 0)$.

Proof. The base cases are immediate:

$$\rho^0 = 1 = 0 \cdot \rho^2 + 0 \cdot \rho + 1, \qquad \rho^1 = 0 \cdot \rho^2 + 1 \cdot \rho + 0, \qquad \rho^2 = 1 \cdot \rho^2 + 0 \cdot \rho + 0.$$

For the inductive step, assume $\rho^n = A_n\rho^2 + B_n\rho + C_n$ with $A_n, B_n, C_n \in \mathbf{Z}$. Then

$$\begin{aligned} \rho^{n+1} &= \rho \cdot \rho^n = A_n\rho^3 + B_n\rho^2 + C_n\rho \\ &= A_n(\rho^2 + \rho + 1) + B_n\rho^2 + C_n\rho \\ &= (A_n + B_n)\rho^2 + (A_n + C_n)\rho + A_n. \end{aligned}$$

Since $A_n, B_n, C_n \in \mathbf{Z}$, the new coefficients $A_n + B_n$, $A_n + C_n$, A_n are also integers. This establishes both the reduction and the recurrence (2). □

Lemma 3.4 (Closure under signed sums)**.** *Any finite signed sum of elements of the form $A\rho^2 + B\rho + C$ with $A, B, C \in \mathbf{Z}$ is again of the same form.*

Proof. $\mathbf{Z}$ is closed under addition and subtraction. Adding or subtracting the integer coefficient triples component-wise yields another integer triple. □

Proof of Theorem 3.2. By Lemma 3.3, each power ρ^{n_j} appearing in the regular CTP expression reduces to an integer triple $(A_{n_j}, B_{n_j}, C_{n_j})$. By Lemma 3.4, the signed sum $\sum \varepsilon_j \rho^{n_j}$ reduces to a single triple $(a, b, c) \in \mathbf{Z}^3$. Uniqueness follows from the basis theorem (Theorem 2.2(iv)). □

Example 3.5.

$$\begin{aligned} \rho^3 &= 1 \cdot \rho^2 + 1 \cdot \rho + 1, \\ \rho^6 + \rho^3 + \rho + 1 &= (7 + 1)\rho^2 + (6 + 1 + 1)\rho + (4 + 1 + 0 + 1) = 8\rho^2 + 8\rho + 6, \\ \rho^6 - \rho^3 - \rho - 1 &= (7 - 1)\rho^2 + (6 - 1 - 1)\rho + (4 - 1 - 0 - 1) = 6\rho^2 + 4\rho + 2. \end{aligned}$$

Remark 3.6. The integer triple (a, b, c) serves as a coordinate system for the regular CTP family. Every member is uniquely labeled by its triple, and no higher powers of ρ are ever needed after reduction.

4 T3: Disguised Cubic Theorem

Theorem 4.1 (Disguised Cubic)**.** *Let $D = a\rho^2 + b\rho + c \in \mathbf{Q}(\rho) \setminus \mathbf{Q}$. Then:*

(i) D satisfies a monic cubic polynomial over $\mathbf{Q}$:

$$m_D(y) = y^3 + Ay^2 + By + C, \qquad A, B, C \in \mathbf{Q}.$$

(ii) If q is any algebraic number satisfying $q^{2^N} = D$, then q is a root of the sparse lifted polynomial

$$x^{3\cdot 2^N} + A\,x^{2^{N+1}} + B\,x^{2^N} + C = 0. \tag{3}$$

Lemma 4.2 (Degree bound for field elements)**.** *Every element of $\mathbf{Q}(\rho)$ has algebraic degree dividing 3 over $\mathbf{Q}$.*

Proof. For any $D \in \mathbf{Q}(\rho)$, the subfield $\mathbf{Q}(D) \subseteq \mathbf{Q}(\rho)$ satisfies

$$[\mathbf{Q}(D) : \mathbf{Q}] \;\Big|\; [\mathbf{Q}(\rho) : \mathbf{Q}] = 3.$$

The divisors of 3 are 1 and 3, so $\deg_{\mathbf{Q}}(D) \in \{1, 3\}$. □

Lemma 4.3 (Non-rational elements are cubic)**.** *If $D \in \mathbf{Q}(\rho) \setminus \mathbf{Q}$, then D has algebraic degree exactly 3 over $\mathbf{Q}$.*

Proof. By Lemma 2.8, $\mathbf{Q}(D) = \mathbf{Q}(\rho)$, so $\deg_{\mathbf{Q}}(D) = [\mathbf{Q}(\rho) : \mathbf{Q}] = 3$. □

Lemma 4.4 (Minimal polynomial via norm)**.** *Let $K = \mathbf{Q}(\rho)$ and $D \in K \setminus \mathbf{Q}$. The minimal polynomial of D over $\mathbf{Q}$ is*

$$m_D(y) = \mathrm{Norm}_{K/\mathbf{Q}}(y - D) = \prod_{i=1}^{3}(y - D_i),$$

where D_1, D_2, D_3 are the conjugates of D under the three embeddings $K \hookrightarrow \mathbf{C}$. Equivalently,

$$m_D(y) = y^3 - \mathrm{Tr}_{K/\mathbf{Q}}(D)\,y^2 + \sigma_2(D)\,y - \mathrm{Norm}_{K/\mathbf{Q}}(D),$$

where $\sigma_2(D) = D_1D_2 + D_1D_3 + D_2D_3$.

Proof. By Lemma 4.3, D has degree 3. The product $\prod(y - D_i)$ is a monic cubic in $\mathbf{Q}[y]$ with D as a root; since D has degree 3, this must be its minimal polynomial. The coefficient formula follows from expanding the product. □

Lemma 4.5 (Substitution lift)**.** *If $m_D(y) = y^3 + Ay^2 + By + C$ and $D = q^{2^N}$, then*

$$q^{3\cdot 2^N} + A\,q^{2^{N+1}} + B\,q^{2^N} + C = 0.$$

Proof. Substitute $y = q^{2^N}$ into $m_D(y) = 0$:

$$\begin{aligned} 0 &= (q^{2^N})^3 + A(q^{2^N})^2 + B(q^{2^N}) + C \\ &= q^{3\cdot 2^N} + A\,q^{2^{N+1}} + B\,q^{2^N} + C. \end{aligned}$$

This is an identity in q. □

Proof of Theorem 4.1. Since $D \in \mathbf{Q}(\rho) \setminus \mathbf{Q}$, Lemma 4.4 gives the minimal polynomial $m_D(y) = \mathrm{Norm}_{K/\mathbf{Q}}(y - D)$, a monic cubic in $\mathbf{Q}[y]$. Setting $y = q^{2^N}$ and applying Lemma 4.5 yields the sparse lifted polynomial (3). □

Example 4.6. Take $D = 8\rho^2 + 8\rho + 6$ (the "3D pure" value). Its minimal polynomial over $\mathbf{Q}$ is

$$m_D(y) = y^3 - 50y^2 + 108y - 88.$$

If $q^8 = D$ (i.e., $N = 3$), then q satisfies

$$x^{24} - 50\,x^{16} + 108\,x^8 - 88 = 0.$$

This polynomial has exactly four terms—powers 24, 16, 8, and 0—which is the hallmark sparse structure predicted by the theorem.

Remark 4.7. The sparse polynomial structure is not accidental. It is forced by the disguised-cubic mechanism: every regular family member, regardless of dimension, inherits its polynomial shape from the underlying cubic minimal polynomial of its field element D, lifted through the substitution $y = x^{2^N}$.

5 T4: Degree Law Theorem

Theorem 5.1 (Degree Law)**.** *Let $D \in \mathbf{Q}(\rho) \setminus \mathbf{Q}$ have algebraic degree 3 over $\mathbf{Q}$. Let q be algebraic with*

$$q^{2^N} = D.$$

Assume the radical extension is full*, i.e.,*

$$[\mathbf{Q}(q) : \mathbf{Q}(D)] = 2^N. \tag{4}$$

Then

$$\deg_{\mathbf{Q}}(q) = 3 \cdot 2^N.$$

Lemma 5.2 (Tower formula)**.** *For a tower of finite field extensions $\mathbf{Q} \subseteq \mathbf{Q}(D) \subseteq \mathbf{Q}(q)$,*

$$[\mathbf{Q}(q) : \mathbf{Q}] = [\mathbf{Q}(q) : \mathbf{Q}(D)] \cdot [\mathbf{Q}(D) : \mathbf{Q}].$$

Proof. This is the standard multiplicativity of degrees in a tower of finite algebraic extensions. If $\{e_1, \ldots, e_m\}$ is a $\mathbf{Q}$-basis for $\mathbf{Q}(D)$ and $\{f_1, \ldots, f_n\}$ is a $\mathbf{Q}(D)$-basis for $\mathbf{Q}(q)$, then $\{e_i f_j\}$ is a $\mathbf{Q}$-basis for $\mathbf{Q}(q)$, giving $[\mathbf{Q}(q) : \mathbf{Q}] = mn$. □

Proof of Theorem 5.1. We apply Lemma 5.2 to the tower $\mathbf{Q} \subseteq \mathbf{Q}(D) \subseteq \mathbf{Q}(q)$. By Lemma 4.3, $[\mathbf{Q}(D) : \mathbf{Q}] = 3$. By hypothesis (4), $[\mathbf{Q}(q) : \mathbf{Q}(D)] = 2^N$. Therefore

$$\deg_{\mathbf{Q}}(q) = [\mathbf{Q}(q) : \mathbf{Q}] = 2^N \cdot 3 = 3 \cdot 2^N.$$

□

Remark 5.3 (On the full-extension hypothesis)*.* Hypothesis (4) asserts that no "degree collapse" occurs when extracting the 2^N-th root. In principle, a collapse would mean that q already belongs to a smaller field, and the polynomial $x^{2^N} - D$ factors non-trivially over $\mathbf{Q}(D)$.

For the pure family members and most subtractive members computed through dimension 6, the sparse lifted polynomials have been verified irreducible over $\mathbf{Q}$, confirming full extension. One known exception is the *2D subtractive* member, whose sparse polynomial

$$x^{12} - 3x^8 - x^4 - 1 = (x^6 - x^4 - x^2 - 1)(x^6 + x^4 - x^2 + 1)$$

factors into two irreducible sextics over $\mathbf{Q}$. The root q satisfies the first factor and has degree $6 = 3 \cdot 2^1$, not $12 = 3 \cdot 2^2$. This is a genuine degree collapse: $[\mathbf{Q}(q) : \mathbf{Q}(D)] = 2$, not 4. The degree of q is still of the form $3 \cdot 2^k$, but with $k = 1$ rather than $k = N = 2$.

A universal no-collapse theorem, or a precise characterization of which family members collapse, is a natural direction for future work. *This question is resolved completely by Theorem 10.2.*

Corollary 5.4 (Degree hierarchy)**.** *Under the full-extension hypothesis, the degrees of the regular family members by dimension are:*

Dimension d	1	2	3	4	5	6
$N = d$	1	2	3	4	5	6
$\deg(q) = 3 \cdot 2^N$	6	12	24	48	96	192

Proof. Direct application of Theorem 5.1 with $N = d$. □

Remark 5.5. The degree sequence $6, 12, 24, 48, 96, 192, \ldots$ is multiplicative and deterministic—each step doubles. This is a structural consequence of the interaction between the fixed cubic base ($\deg = 3$) and the iterated square-root extraction ($\times 2^N$). It is not a numerical coincidence.

6 T5: Structural Surplus Theorem

Definition 6.1 (Structural surplus function)**.** For $b > 0$, define

$$\Delta(b) = \sqrt{b^6 + b^3 + b + 1} - b^3.$$

Theorem 6.2 (Structural Surplus)**.** *(i) For all $b \geq 2$,*

$$\Delta(b) > \frac{1}{2}.$$

(ii) The surplus is asymptotically tight:

$$\lim_{b\to\infty} \Delta(b) = \frac{1}{2}.$$

Lemma 6.3 (Conjugate form)**.** *For all $b > 0$,*

$$\Delta(b) = \frac{b^3 + b + 1}{\sqrt{b^6 + b^3 + b + 1} + b^3}.$$

Proof. Set $X = b^6 + b^3 + b + 1$. Multiply and divide by the conjugate:

$$\Delta(b) = \sqrt{X} - b^3 = \frac{X - b^6}{\sqrt{X} + b^3} = \frac{b^3 + b + 1}{\sqrt{X} + b^3}.$$

□

Lemma 6.4 (Strict lower bound)**.** *For all $b \geq 2$, $\Delta(b) > \frac{1}{2}$.*

Proof. Using Lemma 6.3, $\Delta(b) > \frac{1}{2}$ is equivalent to

$$2(b^3 + b + 1) > \sqrt{b^6 + b^3 + b + 1} + b^3,$$

which rearranges to

$$b^3 + 2b + 2 > \sqrt{b^6 + b^3 + b + 1}.$$

Both sides are positive for $b \geq 2$. Squaring and subtracting:

$$\begin{aligned}(b^3 + 2b + 2)^2 &- (b^6 + b^3 + b + 1)\\ &= b^6 + 4b^4 + 4b^3 + 4b^2 + 8b + 4 - b^6 - b^3 - b - 1\\ &= 4b^4 + 3b^3 + 4b^2 + 7b + 3.\end{aligned}$$

Every term is positive for $b \geq 2$ (indeed for $b \geq 0$), so the difference is strictly positive. Since the squaring step is valid (both sides positive) and the squared inequality is strict, the original inequality $\Delta(b) > \frac{1}{2}$ holds. □

Lemma 6.5 (Asymptotic limit)**.** $\lim_{b\to\infty} \Delta(b) = \frac{1}{2}$.

Proof. Factor b^6 inside the square root:

$$\Delta(b) = b^3\left(\sqrt{1 + b^{-3} + b^{-5} + b^{-6}} - 1\right).$$

Set $u = b^{-3} + b^{-5} + b^{-6} \to 0$ as $b \to \infty$. The Taylor expansion $\sqrt{1+u} = 1 + \frac{u}{2} + O(u^2)$ gives

$$\Delta(b) = b^3\left(\frac{u}{2} + O(u^2)\right) = b^3\left(\frac{b^{-3}}{2} + O(b^{-5})\right) = \frac{1}{2} + O(b^{-2}).$$

Therefore $\Delta(b) \to \frac{1}{2}$. □

Proof of Theorem 6.2. Part (i) is Lemma 6.4. Part (ii) is Lemma 6.5. □

Example 6.6. At $b = 3$:

$$\Delta(3) = \sqrt{3^6 + 3^3 + 3 + 1} - 27 = \sqrt{760} - 27 \approx 0.5680975.$$

This exceeds $\frac{1}{2}$ by approximately 0.068.

Remark 6.7. The surplus function Δ encodes a *constructive capacity*: the gap between the radical of a polynomial in b and the cube b^3. The theorem says this gap is always strictly more than a half-unit, and approaches exactly $\frac{1}{2}$ from above as the base grows. The surplus never vanishes, and its limit is universal.

7 T6: Algebraic P-Family Theorem

Definition 7.1 (Phi-point equation)**.** Let $\varphi = \frac{1+\sqrt{5}}{2}$ denote the golden ratio. For a parameter P, the *phi-point equation* is

$$\varphi^8 = x^8 + x^4 + x^2 + x + P. \tag{5}$$

Theorem 7.2 (Algebraic P-Family)**.** *If P is algebraic (over $\mathbf{Q}$), then every root x of equation (5) is algebraic.*

Lemma 7.3. *The golden ratio φ is algebraic.*

Proof. φ satisfies the polynomial $x^2 - x - 1 = 0$ with integer coefficients. □

Lemma 7.4. *Every polynomial expression in algebraic numbers is algebraic.*

Proof. The set of algebraic numbers, denoted $\overline{\mathbf{Q}}$, forms a field: it is closed under addition, multiplication, and raising to non-negative integer powers. □

Lemma 7.5. *If $g(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$ has all coefficients $a_i \in \overline{\mathbf{Q}}$, then every root of g lies in $\overline{\mathbf{Q}}$.*

Proof. The algebraic closure $\overline{\mathbf{Q}}$ is, by definition, algebraically closed: every polynomial with coefficients in $\overline{\mathbf{Q}}$ splits completely over $\overline{\mathbf{Q}}$. □

Proof of Theorem 7.2. Rewrite equation (5) as

$$x^8 + x^4 + x^2 + x + (P - \varphi^8) = 0.$$

By Lemma 7.3, φ is algebraic. By Lemma 7.4, φ^8 is algebraic. If P is algebraic, then $P - \varphi^8$ is algebraic (since $\overline{\mathbf{Q}}$ is a field). Therefore the polynomial $x^8 + x^4 + x^2 + x + (P - \varphi^8)$ has algebraic coefficients, and by Lemma 7.5 every root is algebraic. □

Example 7.6.
- $P = 1$ yields the degree-16 phi-point member (Theorem 8.1).
- $P = 1/2$, $P = 1/\varphi$, $P = 1/\rho$ all produce algebraic roots.
- If one demanded a transcendental output from this equation, the corresponding P would necessarily be transcendental.

Remark 7.7. This theorem shows that the phi-point equation (5) defines an algebraic parameter family: as P ranges over $\overline{\mathbf{Q}}$, the roots remain algebraic. The family sits alongside the regular CTP family as an independent algebraic structure.

8 T7: Phi-Point Polynomial Theorem

Theorem 8.1 (Phi-Point Polynomial)**.** *For $P = 1$, the phi-point equation*

$$\varphi^8 = x^8 + x^4 + x^2 + x + 1$$

defines a real algebraic root x satisfying the degree-16 polynomial

$$\begin{aligned} &x^{16} + 2x^{12} + 2x^{10} + 2x^9 - 44x^8 \\ &\quad + 2x^6 + 2x^5 - 44x^4 + 2x^3 - 44x^2 - 45x - 45 = 0. \end{aligned} \tag{6}$$

Proof. **Step 1: Reduce φ^8 in the golden-ratio field.** Since $\varphi^2 = \varphi + 1$, repeated squaring gives

$$\begin{aligned} &\varphi^2 = \varphi + 1, \\ &\varphi^4 = (\varphi + 1)^2 = \varphi^2 + 2\varphi + 1 = 3\varphi + 2, \\ &\varphi^8 = (3\varphi + 2)^2 = 9\varphi^2 + 12\varphi + 4 = 9(\varphi + 1) + 12\varphi + 4 = 21\varphi + 13. \end{aligned}$$

Step 2: Express φ in terms of x. From $\varphi^8 = x^8 + x^4 + x^2 + x + 1$ and $\varphi^8 = 21\varphi + 13$:

$$21\varphi + 13 = x^8 + x^4 + x^2 + x + 1,$$

so

$$\varphi = \frac{x^8 + x^4 + x^2 + x - 12}{21}.$$

Step 3: Eliminate φ via its minimal polynomial. Substitute into $\varphi^2 - \varphi - 1 = 0$ and multiply by $21^2 = 441$:

$$(x^8 + x^4 + x^2 + x - 12)^2 - 21(x^8 + x^4 + x^2 + x - 12) - 441 = 0.$$

Let $u = x^8 + x^4 + x^2 + x - 12$. Then $u^2 - 21u - 441 = 0$ becomes

$$(x^8 + x^4 + x^2 + x - 12)^2 - 21(x^8 + x^4 + x^2 + x - 12) - 441 = 0.$$

Expanding $(x^8 + x^4 + x^2 + x - 12)^2$:

$$\begin{aligned} &x^{16} + 2x^{12} + 2x^{10} + 2x^9 - 24x^8 + x^8 + 2x^6 + 2x^5 \\ &\quad - 24x^4 + x^4 + 2x^3 - 24x^2 + x^2 - 24x + 144 \\ &= x^{16} + 2x^{12} + 2x^{10} + 2x^9 - 23x^8 + 2x^6 + 2x^5 \\ &\quad - 23x^4 + 2x^3 - 23x^2 - 24x + 144. \end{aligned}$$

Subtracting $21(x^8 + x^4 + x^2 + x - 12)$:

$$\begin{aligned} &x^{16} + 2x^{12} + 2x^{10} + 2x^9 - 23x^8 + 2x^6 + 2x^5 - 23x^4 + 2x^3 - 23x^2 - 24x + 144 \\ &\quad - 21x^8 - 21x^4 - 21x^2 - 21x + 252. \end{aligned}$$

Subtracting 441:

$$\begin{aligned} &x^{16} + 2x^{12} + 2x^{10} + 2x^9 - 44x^8 + 2x^6 + 2x^5 \\ &\quad - 44x^4 + 2x^3 - 44x^2 - 45x - 45 = 0. \end{aligned}$$

This is precisely the polynomial (6). □

Remark 8.2 (Numerical verification). The real root of (6) near $\pi/2$ is

$$x \approx 1.5649082789076842913868550165291812766.$$

Doubling:

$$2x \approx 3.1298165578153685827737100330583625532.$$

Remark 8.3. This theorem is purely algebraic. The proximity of $2x$ to π is a numerical observation, not a theorem-level claim.

Proposition 8.4 (Irreducibility of the phi-point polynomial)**.** *The polynomial* (6) *is irreducible over* $\mathbf{Q}$*. Therefore the real root* x *has algebraic degree exactly* 16*.*

Proof. The polynomial (6) has integer coefficients and is monic. Reducing modulo $p = 13$, one verifies that the resulting polynomial in $\mathbf{F}_{13}[x]$ has no factors of degree less than 16—that is, it is irreducible over $\mathbf{F}_{13}$. By the classical criterion (a monic integer polynomial that is irreducible modulo a prime p is irreducible over $\mathbf{Q}$), the polynomial is irreducible. □

9 T8: Kappa-Rho Theorem

Definition 9.1. Define

$$S = \rho^{10} + \rho^6 + \rho^3 + \rho + 1$$

and the *kappa-rho constant*

$$\kappa_\rho = \sqrt{\sqrt{S}} - \rho^2 = S^{1/4} - \rho^2.$$

Theorem 9.2 (Kappa-Rho)**.** *The constant κ_ρ is algebraic and satisfies the degree-12 polynomial*

$$\begin{aligned} &x^{12} + 12x^{11} + 50x^{10} + 76x^9 - 478x^8 + 752x^7 \\ &\quad - 1000x^6 + 720x^5 + 1120x^4 - 1488x^3 - 560x^2 + 912x - 676 = 0. \end{aligned} \tag{7}$$

Lemma 9.3. *The sum $S = \rho^{10} + \rho^6 + \rho^3 + \rho + 1$ lies in $\mathbf{Q}(\rho)$.*

Proof. Each term $\rho^{10}, \rho^6, \rho^3, \rho, 1$ is an element of $\mathbf{Q}(\rho)$, and $\mathbf{Q}(\rho)$ is closed under addition. By the reduction relation (Theorem 2.2), S reduces to a unique element $a\rho^2 + b\rho + c$ with $a, b, c \in \mathbf{Z}$.

Explicitly, using the recurrence from Lemma 3.3:

$$\begin{aligned} \rho^3 &= \rho^2 + \rho + 1, \\ \rho^6 &= 7\rho^2 + 6\rho + 4, \\ \rho^{10} &= 81\rho^2 + 68\rho + 44. \end{aligned}$$

Therefore

$$S = (81 + 7 + 1)\rho^2 + (68 + 6 + 1 + 1)\rho + (44 + 4 + 1 + 0 + 1) = 89\rho^2 + 76\rho + 50.$$

□

Lemma 9.4. *The quantity $u = S^{1/4} = \sqrt{\sqrt{S}}$ is algebraic.*

Proof. Since $S \in \mathbf{Q}(\rho) \subset \overline{\mathbf{Q}}$, the number S is algebraic. Taking a fourth root of an algebraic number yields an algebraic number (it is a root of $y^4 - S$, which has algebraic coefficients; apply Lemma 7.5). □

Lemma 9.5. *Subtracting ρ^2 from an algebraic number yields an algebraic number.*

Proof. ρ^2 is algebraic (as an element of $\mathbf{Q}(\rho)$), and $\overline{\mathbf{Q}}$ is closed under subtraction. □

Proof of algebraicity. By Lemma 9.3, S is algebraic. By Lemma 9.4, $u = S^{1/4}$ is algebraic. By Lemma 9.5, $\kappa_\rho = u - \rho^2$ is algebraic. □

Derivation of the polynomial. Set $\kappa_\rho = x$ and $u = x + \rho^2$, so $u^4 = S$. Then

$$(x + \rho^2)^4 = 89\rho^2 + 76\rho + 50.$$

This is a polynomial equation in two algebraic quantities, x and ρ. To eliminate ρ, introduce the auxiliary variable $t = \rho$ and form the two-variable system:

$$\begin{aligned} g(x, t) &= (x + t^2)^4 - 89t^2 - 76t - 50 = 0, \\ h(t) &= t^3 - t^2 - t - 1 = 0. \end{aligned}$$

The resultant $\mathrm{Res}_t\big(g(x,t),\ h(t)\big)$ eliminates t and produces a polynomial in x alone. Direct computation of this resultant yields exactly the degree-12 polynomial (7), with no extraneous factors. □

Proposition 9.6 (Irreducibility of the kappa-rho polynomial)**.** *The polynomial* (7) *is irreducible over* $\mathbf{Q}$*. Therefore* $\deg(\kappa_\rho) = 12$*.*

Proof. The polynomial has integer coefficients and is monic. Reducing modulo $p = 5$, one verifies that the resulting polynomial in $\mathbf{F}_5[x]$ is irreducible (it has no factors of degree less than 12). By the mod-p irreducibility criterion, the polynomial is irreducible over $\mathbf{Q}$. □

Remark 9.7 (Numerical verification and irreducibility)*.* The numerical value

$$\kappa_\rho \approx 1.32399784289355113114890140612855889 43$$

substituted into (7) yields a residual consistent with exact satisfaction (zero to full working precision). By Proposition 9.6, the polynomial is the minimal polynomial of κ_ρ, confirming $\deg(\kappa_\rho) = 12$.

Remark 9.8 (Exceptional status)*.* The degree of κ_ρ is 12, which does *not* fit the regular family pattern $3 \cdot 2^N$. The regular pattern would give degrees $6, 12, 24, 48, \ldots$, and while $12 = 3 \cdot 2^2$ matches numerically, κ_ρ is not generated by the regular 2^N-th root mechanism. It arises from a fourth root of a field element followed by a shift by ρ^2—a fundamentally different construction. This makes κ_ρ a genuine *exceptional* algebraic constant within the CTP framework: it demonstrates that the framework contains algebraic structure beyond the regular branch.

10 T9: Radical Collapse Criterion

Theorem 5.1 established the degree law $\deg(q) = 3 \cdot 2^N$ under the hypothesis that the radical extension $[\mathbf{Q}(q) : \mathbf{Q}(D)] = 2^N$ is full. This section resolves that hypothesis completely by providing an exact, checkable algebraic criterion for when collapse occurs.

Definition 10.1 (Radical depth)**.** For $D \in \mathbf{Q}(\rho)$ with $D > 0$, define the *radical depth* of D in $\mathbf{Q}(\rho)$ as

$$k(D) = \max\{\, j \geq 0 : D^{1/2^j} \in \mathbf{Q}(\rho) \,\}.$$

Equivalently, $k(D)$ is the largest integer such that D is a perfect 2^k-th power of an element of $\mathbf{Q}(\rho)$.

Theorem 10.2 (Radical Collapse Criterion)**.** *Let $D \in \mathbf{Q}(\rho) \setminus \mathbf{Q}$ with $D > 0$, and let q be the positive real number satisfying $q^{2^N} = D$. Then:*

(i) The algebraic degree of q over $\mathbf{Q}$ is exactly

$$\deg(q) = 3 \cdot 2^{\max(N-k(D),\, 0)}. \tag{8}$$

When $k(D) \geq N$, the root $q = D^{1/2^N}$ already lies in $\mathbf{Q}(\rho)$ and $\deg(q) = 3$.

(ii) In particular, the full-extension hypothesis of Theorem 5.1 holds if and only if $k(D) = 0$—that is, if and only if D is not a perfect square in $\mathbf{Q}(\rho)$.

Lemma 10.3 (Capelli's irreducibility criterion, 1898)**.** *Let K be a field of characteristic zero, and let $a \in K$ with $a > 0$. Then the polynomial $x^{2^n} - a$ is irreducible over K if and only if a is not a perfect square in K.*

Proof. This is a special case of the Vahlen–Capelli theorem: for n a prime power p^k, the polynomial $x^{p^k} - a$ is irreducible over K if and only if a is not a p-th power in K, with an additional condition when $p = 2$ and $k \geq 2$ that $a \neq -4b^4$ for any $b \in K$.

For $p = 2$ and $a > 0$: the condition $a \neq -4b^4$ is automatically satisfied since $-4b^4 \leq 0 < a$. Therefore $x^{2^n} - a$ is irreducible over K if and only if a is not a perfect square in K. □

Proof of Theorem 10.2. Set $k = k(D)$. By definition, $D = \alpha^{2^k}$ for some $\alpha \in \mathbf{Q}(\rho)$ with $\alpha > 0$. By maximality of k, the element α is not a square in $\mathbf{Q}(\rho)$.

Case 1: $N \leq k$**.** Then $q = D^{1/2^N} = \alpha^{2^{k-N}} \in \mathbf{Q}(\rho)$, and $q \notin \mathbf{Q}$ (since $q^{2^N} = D \notin \mathbf{Q}$). Therefore $\deg(q) = [\mathbf{Q}(\rho) : \mathbf{Q}] = 3 = 3 \cdot 2^0 = 3 \cdot 2^{\max(N-k,\,0)}$.

Case 2: $N > k$**.** Then $q^{2^{N-k}} = \alpha$. Since α is not a square in $\mathbf{Q}(\rho)$, Lemma 10.3 shows that $x^{2^{N-k}} - \alpha$ is irreducible over $\mathbf{Q}(\rho)$. Therefore $[\mathbf{Q}(q) : \mathbf{Q}(\rho)] = 2^{N-k}$, and by the tower law,

$$\deg(q) = [\mathbf{Q}(q) : \mathbf{Q}] = [\mathbf{Q}(q) : \mathbf{Q}(\rho)] \cdot [\mathbf{Q}(\rho) : \mathbf{Q}] = 2^{N-k} \cdot 3 = 3 \cdot 2^{N-k} = 3 \cdot 2^{\max(N-k,\,0)}.$$

□

Remark 10.4. The max is necessary: for example, $D = \rho^4 = 2\rho^2 + 2\rho + 1$ has $k(D) = 2$ (since $\rho^4 = (\rho^2)^2$ and $\rho^2 = \rho^2$, but ρ is not a square in $\mathbf{Q}(\rho)$). Taking $N = 1$, the root $q^2 = \rho^4$ gives $q = \rho^2 \in \mathbf{Q}(\rho)$, so $\deg(q) = 3 = 3 \cdot 2^{\max(1-2,0)} = 3 \cdot 2^0$. Without the max, the formula would yield $3 \cdot 2^{-1} = 3/2$, which is impossible.

Remark 10.5 (Verification against computed family). Theorem 10.2 correctly predicts the degree of every computed family member:

Member	D **as triple**	N	$k(D)$	$\deg(q)$	**Status**
1D pure	$(0,1,1)$	1	0	6	full extension
1D sub	$(0,1,-1)$	1	0	6	full extension
2D pure	$(1,2,2)$	2	0	12	full extension
2D sub	$(1,0,0)=\rho^2$	2	1	6	collapse ($\sqrt{D}=\rho$)
3D pure	$(8,8,6)$	3	0	24	full extension
3D sub	$(6,4,2)$	3	0	24	full extension
4D sub	$(73,60,38)$	4	0	48	full extension

The 2D subtractive member is the only collapse: its field element $D=\rho^2$ is a perfect square in $\mathbf{Q}(\rho)$ (with $\sqrt{D}=\rho$), so the root $q=\rho^{1/2}$ requires only a single square-root extension, not a double. Every other computed member has $k(D)=0$.

10.1 Structural consequence: cubic fields resist collapse

Lemma 10.6 (Norm obstruction)**.** *Let $K=\mathbf{Q}(\rho)$. If $D\in K$ is a perfect square in K, then $\mathrm{Norm}_{K/\mathbf{Q}}(D)$ is a perfect square in $\mathbf{Q}$.*

Proof. If $D=\alpha^2$ with $\alpha\in K$, then $\mathrm{Norm}(D)=\mathrm{Norm}(\alpha^2)=\mathrm{Norm}(\alpha)^2$, which is a rational square. □

Remark 10.7. This gives a fast necessary condition for collapse. The field norm of D is the constant term (up to sign) of its minimal polynomial: if $m_D(y)=y^3+Ay^2+By+C$, then $\mathrm{Norm}(D)=-C$. Among the computed CTP members:

Member	$\mathrm{Norm}(D)$	**Square?**
1D pure / sub	2	no → no collapse
2D pure	13	no → no collapse
2D sub	1	yes → test inconclusive
3D pure	88	no → no collapse
3D sub	32	no → no collapse
4D sub	763	no → no collapse

The norm obstruction alone rules out collapse for six of seven members. Only the 2D subtractive case ($\mathrm{Norm}=1=1^2$) passes the test—and it is indeed the sole collapse.

The following corollary gives precise algebraic meaning to the dimensional distinction between quadratic and cubic fields.

Corollary 10.8 (Quadratic vs. cubic squareness)**.** *Let φ denote the golden ratio (root of x^2-x-1), and let ρ denote the tribonacci constant (root of x^3-x^2-x-1).*

(i) In the quadratic field $\mathbf{Q}(\varphi)$: every even power φ^{2m} is a perfect square in $\mathbf{Q}(\varphi)$, since $\varphi^{2m}=(\varphi^m)^2$ and $\varphi^m\in\mathbf{Q}(\varphi)$. The corresponding sparse polynomials always factor.

(ii) In the cubic field $\mathbf{Q}(\rho)$: an element $D=a\rho^2+b\rho+c$ is a perfect square in $\mathbf{Q}(\rho)$ if and only if the nonlinear system

$$(p\rho^2+q\rho+r)^2=a\rho^2+b\rho+c \pmod{\rho^3-\rho^2-\rho-1} \tag{9}$$

has a solution $(p, q, r) \in \mathbf{Q}^3$. This is a system of three quadratic equations in three unknowns, and constitutes a nontrivial Diophantine constraint. Among the seven computed CTP family members, only the triple $(1, 0, 0)$ (i.e., $D = \rho^2$) admits a solution.

Proof. **(i)** $\varphi^m \in \mathbf{Q}(\varphi)$ for all $m \geq 0$ (since $\mathbf{Q}(\varphi)$ is closed under multiplication). Therefore $\varphi^{2m} = (\varphi^m)^2$ is a perfect square in $\mathbf{Q}(\varphi)$.

Concretely: the sparse polynomials for $D = \varphi^2, \varphi^4, \varphi^8, \varphi^{16}$ all factor over $\mathbf{Q}$, and each factorization is cumulative (each higher power acquires one additional irreducible factor).

(ii) Expanding (9) and reducing modulo $\rho^3 = \rho^2 + \rho + 1$ yields:

$$\begin{aligned} \rho^2 : &\quad 2p^2 + 2pq + 2pr + q^2 = a, \\ \rho^1 : &\quad 2p^2 + 2pq + 2qr = b, \\ \rho^0 : &\quad p^2 + 2pq + r^2 = c. \end{aligned}$$

This is a system of three quadratic equations in three unknowns (p, q, r). For the specific triple $(1, 0, 0)$ (i.e., $D = \rho^2$), the solution $(p, q, r) = (0, 1, 0)$ exists: $\rho^2 = \rho \cdot \rho$.

Among the seven computed CTP family members, only $(1, 0, 0)$ admits a solution. □

Remark 10.9 (Structural observation). The contrast between parts (i) and (ii) is noteworthy. In the quadratic field, the squareness of even generator powers is automatic—a one-line identity. In the cubic field, squareness requires solving a system of three coupled Diophantine equations, and computationally the solution set among CTP family members is sparse (one out of seven).

This structural asymmetry—that squareness is trivial in degree 2 but constrained in degree 3—provides an algebraic counterpart to the motivational distinction drawn in the prefaces. It does not constitute a proof that collapse is "rare" in any measure-theoretic sense, but it does show that the cubic field imposes a concrete algebraic barrier to collapse that has no quadratic analog.

11 Corollaries and Dimension-Specific Instances

Corollary 11.1 (Explicit sparse polynomials)**.** *Every regular family member already computed is an explicit root of an integer-coefficient sparse polynomial. The known instances are:*

Member	***Degree***	***Sparse polynomial***
*0D seed**	*1*	$x - 1 = 0$
1D pure	*6*	$x^6 - 4x^4 + 4x^2 - 2 = 0$
1D subtractive	*6*	$x^6 + 2x^4 - 2 = 0$
2D pure	*12*	$x^{12} - 11x^8 + 19x^4 - 13 = 0$
2D subtractive	$6^\dagger$	$x^{12} - 3x^8 - x^4 - 1 = 0$
3D pure	*24*	$x^{24} - 50x^{16} + 108x^8 - 88 = 0$
3D subtractive	*24*	$x^{24} - 28x^{16} - 48x^8 - 32 = 0$
4D subtractive	*48*	$x^{48} - 393x^{32} - 913x^{16} - 763 = 0$
5D pure	*96*	$x^{96} - 9820x^{64} + 20976x^{32} - 17486 = 0$
5D subtractive	*96*	$x^{96} - 8834x^{64} - 18752x^{32} - 15334 = 0$

Higher-dimensional instances (6D and beyond) follow the same pattern.

†*Degree collapse. The 2D subtractive sparse polynomial factors over* $\mathbf{Q}$ *as* $(x^6 - x^4 - x^2 - 1)(x^6 + x^4 - x^2 + 1)$*. The root* q *satisfies the first (irreducible) factor, so* $\deg(q) = 6$*, not* 12*. This is the only known collapse among the computed family members.*

**Degenerate case. The 0D seed has* $D = 1 \in \mathbf{Q}$*, so Theorems T3 and T4 do not apply. It is listed for completeness as the trivial base case of the family.*

Corollary 11.2 (Degree doubling)**.** *The regular family exhibits the degree sequence*

$$6,\ 12,\ 24,\ 48,\ 96,\ 192,\ \ldots$$

corresponding to dimensions $d = 1, 2, 3, 4, 5, 6, \ldots$ *under the full-extension hypothesis.*

Corollary 11.3 (Explicit surplus at $b = 3$)**.** *The structural surplus at base* $b = 3$ *is*

$$\Delta(3) = \sqrt{760} - 27 \approx 0.5680975,$$

which satisfies $\Delta(3) > \frac{1}{2}$*.*

Corollary 11.4 (Independence of the phi-point family)**.** *The phi-point family (Theorems 7.2–8.1) provides a separate algebraic branch not generated by the regular CTP reductions. It is rooted in the golden ratio rather than the tribonacci constant.*

12 Summary of Mathematical Content

This packet establishes six main results about the algebraic structure of the Christopherson Tribonacci Projection:

1. **A real algebraic family exists** (T1–T3). The CTP framework produces a coherent algebraic family structured by a fixed cubic base field $\mathbf{Q}(\rho)$, universal reduction to integer-coefficient quadratics in ρ, sparse lifted polynomials, and a dimension-linked degree hierarchy. This family is systematic—not ad hoc.

2. **The degree law is complete** (T4 + T9). The algebraic degree of each family member is exactly $\deg(q) = 3 \cdot 2^{\max(N-k(D),0)}$, where $k(D)$ is the radical depth of the field element D in $\mathbf{Q}(\rho)$. When $k(D) = 0$ (the typical case), this gives the full degree law $\deg(q) = 3 \cdot 2^N$. The sole known exception among CTP family members is the 2D subtractive member ($D = \rho^2$, $k = 1$, degree 6 instead of 12). The criterion is checkable: $k(D) = 0$ if and only if D is not a perfect square in $\mathbf{Q}(\rho)$.

3. **The structural surplus is strictly positive** (T5). The constructive-capacity function $\Delta(b) = \sqrt{b^6 + b^3 + b + 1} - b^3$ is bounded below by $\frac{1}{2}$ for all $b \geq 2$, with $\lim_{b\to\infty} \Delta(b) = \frac{1}{2}$. The surplus never vanishes.

4. **A second algebraic family exists** (T6–T7). The phi-point equation defines an independent algebraic parameter family: for every algebraic P, the roots are algebraic. The $P = 1$ member satisfies an explicit degree-16 polynomial.

5. **An exceptional constant lies outside the regular branch** (T8). The kappa-rho constant κ_ρ is algebraic of degree exactly 12 (its polynomial is verified irreducible over $\mathbf{Q}$) and is not generated by the regular 2^N-root mechanism. It represents genuine exceptional structure in the framework.

6. **The cubic field imposes an algebraic barrier to collapse that the quadratic field does not** (T9). In the quadratic field $\mathbf{Q}(\varphi)$, every even power of the generator is automatically a perfect square, and radical towers built from such elements collapse at every level. In the cubic field $\mathbf{Q}(\rho)$, squareness requires solving a system of three coupled Diophantine equations—a nontrivial constraint. Among all computed CTP family members, only one ($D = \rho^2$) satisfies it.

   This structural asymmetry provides an algebraic counterpart to the dimensional distinction drawn in the prefaces: constructions rooted in a degree-2 ground field are vulnerable to collapse via trivial square-root identities; constructions rooted in a degree-3 ground field face a concrete algebraic barrier. The degree law $\deg(q) = 3 \cdot 2^N$ holds whenever that barrier is not overcome.

End of theorem packet.